

RAFFLES INSTITUTION
2017 Year 6 Preliminary Examination

Higher 2

BIOLOGY

9744/01

Paper 1 Multiple Choice

26th September 2017

1 hour

Additional Materials: Multiple Choice Answer Sheet

READ THESE INSTRUCTIONS FIRST

Write in soft pencil.

Do not use staples, paper clips, glue or correction fluid.

Write your name and shade your index number on the Answer Sheet in the spaces provided unless this has been done for you.

There are **thirty** questions in this paper. Answer all questions. For each question there are four possible answers **A, B, C, and D**.

Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

Read the instructions on the Answer Sheet very carefully.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.

Any rough working should be done in this booklet.

The use of an approved scientific calculator is expected, where appropriate.

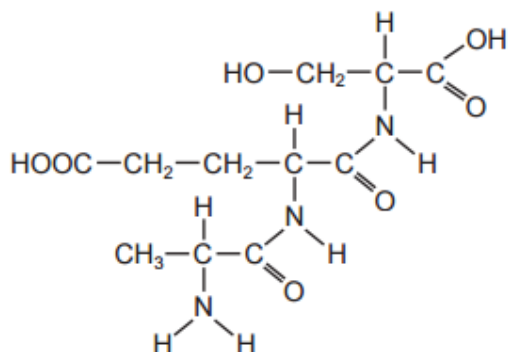
This document consists of **20** printed pages.



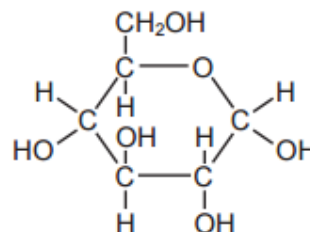
Raffles Institution
Internal Examination

1. Sample A contains a mixture of biological molecules. It was tested using Benedict's reagent, biuret solution and ethanol. After testing, the solutions were blue with the Benedict's test, purple with the biuret test and cloudy with ethanol emulsion test.

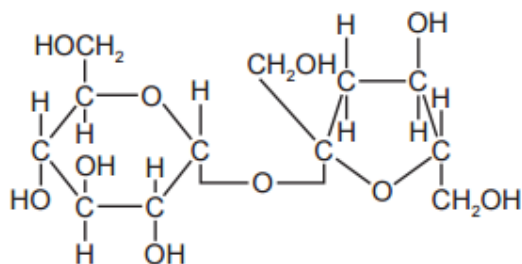
1



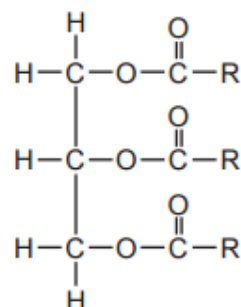
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3



4



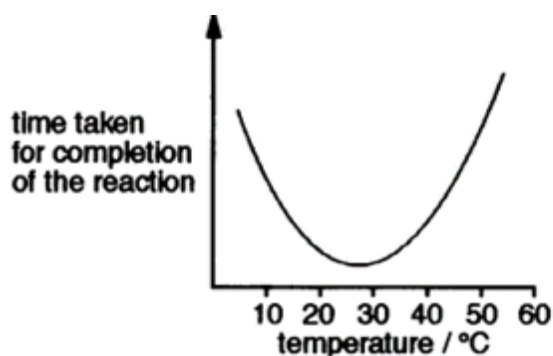
Which molecules could the mixture contain?

- A 1 only
- B 3 and 4 only
- C 1, 3 and 4 only
- D 2, 3 and 4 only

2. A fixed concentration of this enzyme was added to a fixed concentration of its substrate. The time taken for completion of the reaction was measured at different temperatures from 5°C to 55°C. An enzyme was found to be completely denatured at 50°C.

Which of the following is true?

- I The graph below represents the above reaction.



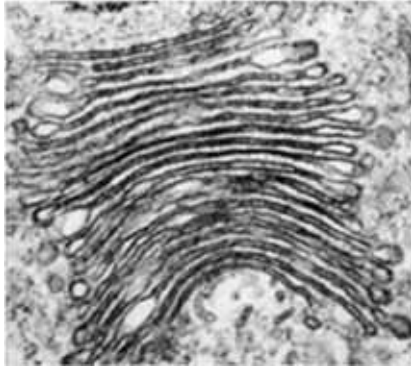
- II As the temperature increases, the activity of the enzymes increases.
III The optimum temperature of the enzyme is less than 50°C.
IV The kinetic energy of the molecules was highest at 50°C.

- A I and III only
B I and II only
C II and IV only
D III only

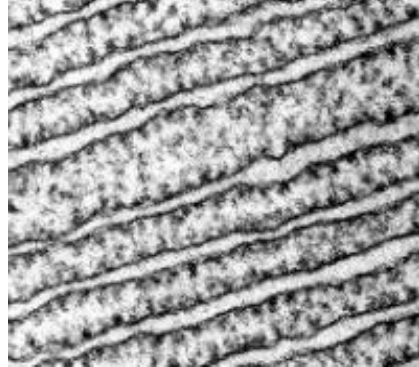
3. As part of a medical check-up, a doctor decided to conduct a liver biopsy for his patient who is a chronic alcoholic. After the patient's liver cells had been extracted and examined. It was compared to a normal cell.

Which of the following organelles could he expect to see in higher quantities?

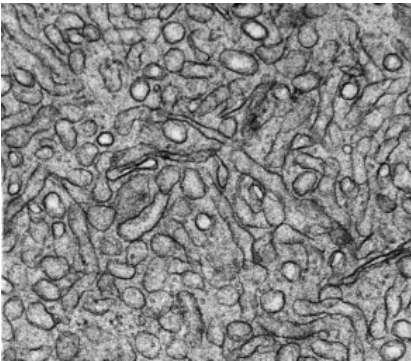
A



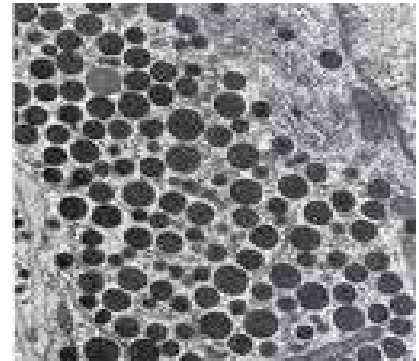
B



C



D



4. Beetroot cells contain a water-soluble red pigment. Two test tubes were set up as described in the table.

tube 1	Pieces of washed raw beetroot in water
tube 2	Pieces of washed raw beetroot in water containing 3 drops of cyanide, a respiratory inhibitor.

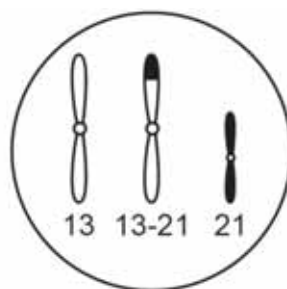
After 30 minutes, the water in tube 2 contained a red pigment but the water in tube 1 did not.

Which of the following statements are false for tube 2?

- I Pigment molecules passed out and were replaced by cyanide.
 - II The cell membrane was unable to retain the red pigment.
 - III Water entered the tissue by osmosis and caused the cells to burst.
 - IV Water passed out of the cells by osmosis and carried the soluble pigment with it.
 - V The same result will occur if ethanol was used instead of cyanide.
-
- A I and III only
 - B III and IV only
 - C II and V only
 - D I, III and IV only

5. Down's syndrome can be caused by a trisomy of chromosome 21, but can also result from translocation of chromosome 21 onto chromosome 13, forming a single chromosome 13-21.

The diagram below shows chromosomes 13 and 21 in the nucleus of a diploid ($2n$) testis cell from a phenotypically normal male carrier of a 13-21 translocation. This cell has a chromosome number of 45.



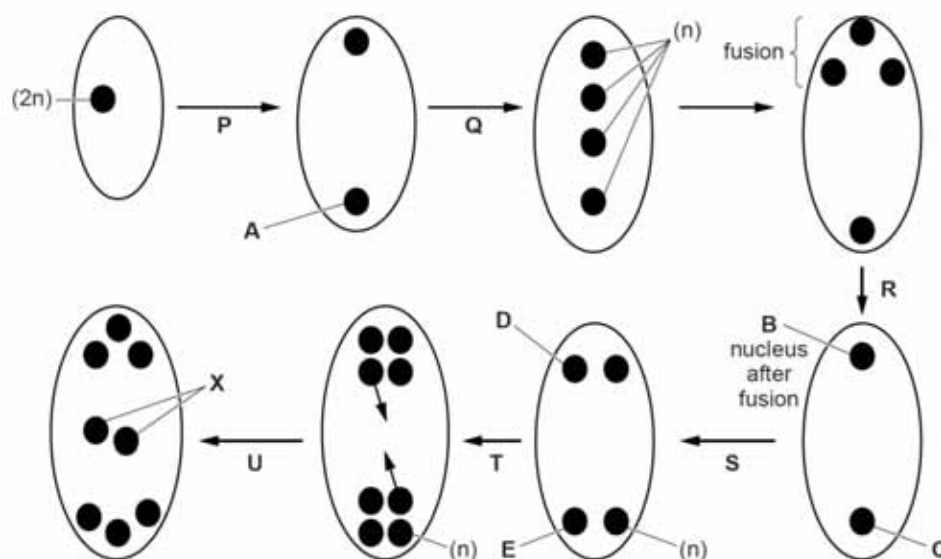
Which is not a likely outcome of fertilisation of normal oocytes by sperm from this male?

	chromosomes in sperm	embryo
A	13 and 21	$2n = 46$ normal phenotype
B	13-21	$2n = 45$ normal phenotype
C	13-21 and 21	$2n = 46$ Down's syndrome
D	13-21 and 21	$2n = 47$ Down's syndrome

6. The development of the embryo sac in flowering plants involves both mitosis and meiosis. Details of this development can vary in different plants.

The diagrams summarise the development of the egg cell within the embryo sac of *Lilium* sp.

Some of the nuclei have been labelled to indicate the ploidy: n = haploid; $2n$ = diploid.

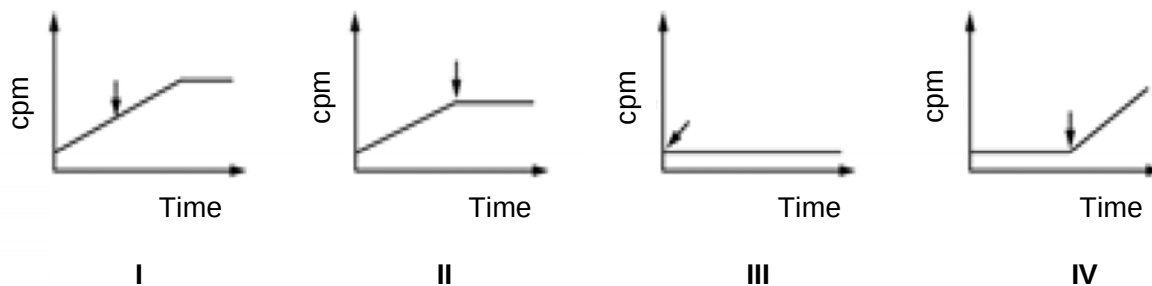


Choose the option that correctly matches mitosis and meiosis to P, Q, S and T.

	P	Q	S	T
A	mitosis	meiosis 1	meiosis 2	mitosis
B	meiosis 1	meiosis 2	mitosis	mitosis
C	mitosis	meiosis 2	mitosis	meiosis 1
D	meiosis 1	meiosis 2	mitosis	meiosis 1

7. The graphs below represent the various curves obtained in a project to study the effect of antibiotics on bacteria.

Radioactively-labelled amino acids were added to bacterial cultures. Various antibiotics, represented by arrows, were added at different times. The peptide products obtained were analysed as counts per minute (cpm).



Which of graph(s) represent the addition of an antibiotic that binds irreversibly to the A site of the ribosome?

- A III only
- B IV only
- C I and IV only
- D II and III only

8. Use the table to answer the following question.

Three-letter codons of messenger RNA and the amino acids specified by the codons			
AAU } Asparagine AAC }	CAU } Histidine CAC }	GAU } Aspartic acid GAC }	UAU } Tyrosine UAC }
AAA } Lysine AAG }	CAA } Glutamine CAG }	GAA } Glutamate GAG }	UAA } Stop UAG }
ACU } ACC } Threonine ACA } ACG }	CCU } CCC } Proline CCA } CCG }	GCU } GCC } Alanine GCA } GCG }	UCU } UCC } Serine UCA } UCG }
AGU } Serine AGC }	CGU } CGC } Arginine CGA } CGG }	GGU } GGC } Glycine GGA } GGG }	UGU } Cysteine UGC }
AGA } Arginine AGG }			UGA – Stop UGG – Tryptophan
AUU } AUC } Isoleucine AUA }	CUU } CUC } Leucine CUA } CUG }	GUU } GUC } Valine GUA } GUG }	UUU } Phenylalanine UUC }
AUG – Methionine			UUA } Leucine UUG }

A cell grown in the presence of radioactively-labelled uracil is actively producing a protein containing lysine.

Which of the following would be radioactive?

- A The amino acid lysine.
- B The codons for lysine.
- C The anticodons for lysine.
- D The DNA code for lysine.

9. The following table shows the genome size, number of genes and chromosome number for a variety of organisms.

organism	genome size (kilo bp)	number of genes	chromosome number
<i>E. coli</i>	4,000	4,000	$n = 1$
Yeast	12,000	6,000	$2n = 12$
Amoeba	290,000,000	No data	500-1000 (possibly polyploid)
Mouse	3,000,000	No data	$2n = 64$
Rhesus monkey	3,000,000	No data	$2n = 42$
Fruit fly	137,000	14,000	$2n = 8$
Humans	3,000,000	30,000	$2n = 46$

From this data it is possible to conclude that

- A there is more non-coding DNA in humans than in the bacterium.
 - B as chromosome number increases, so do the number of genes.
 - C the mouse and the rhesus monkey will have the same number of genes.
 - D the genome size relates to the complexity of an organism.
10. Transcription factors in eukaryotes must be able to perform the following functions except
- A transcribe different classes of RNA.
 - B enter and leave the nucleus.
 - C interact with RNA polymerase.
 - D bind to DNA.

11. Which of the following statement(s) regarding viruses is/are incorrect?

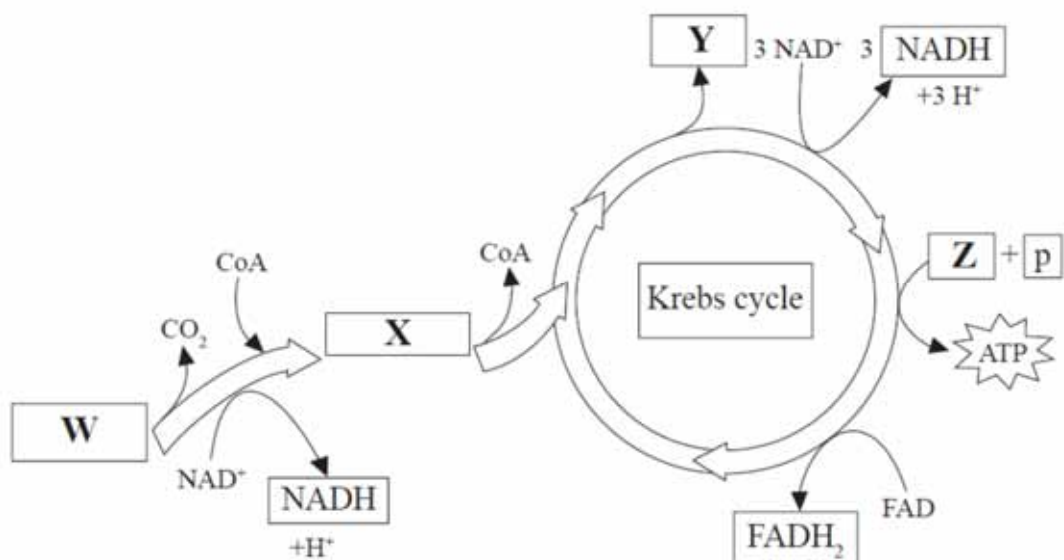
- I When viruses go through antigenic drift, two different viruses may have infected a single host cell and recombined into a new virus.
 - II The DNA-dependent RNA polymerases that are required for the replication of influenza viral genome in the host cell are of viral origin.
 - III Cytotoxic T cells can kill virus-infected target cells by releasing perforins that create pores in the infected cell and lysozymes that activate enzymes that trigger apoptosis of the cell respectively.
 - IV The enzyme integrase is involved in the integration of viral DNA into the host cell genome in both the lambda phage and human immunodeficiency virus life cycles.
 - V For the influenza virus to enter the host cell, haemagglutinin on the host cell membrane binds to a sialic acid receptor of the virus.
- A IV only
 - B I, II and IV only
 - C I, II, III and V only
 - D All of the above

12. Which of the following statement(s) is/are true?

- I A mutation that inactivates the regulatory gene of a repressible operon in *E.coli* would result in the continuous transcription of the structural genes that are controlled by that regulator.
 - II During generalised transduction, the prophage may be improperly excised to include the adjacent segment of bacterial DNA during an induction event.
 - III A mutation in the regulatory gene that resulted in the permanent repression of the *lac* operon could be due to the inability of the mutant repressor to bind to the inducer.
 - IV The F factor on the F plasmid, codes for proteins necessary for the formation of sex pili and subsequent cytoplasmic mating bridge, allowing for conjugation to occur between bacteria.
- A III only
 - B I and II only
 - C I, III and IV only
 - D All of the above

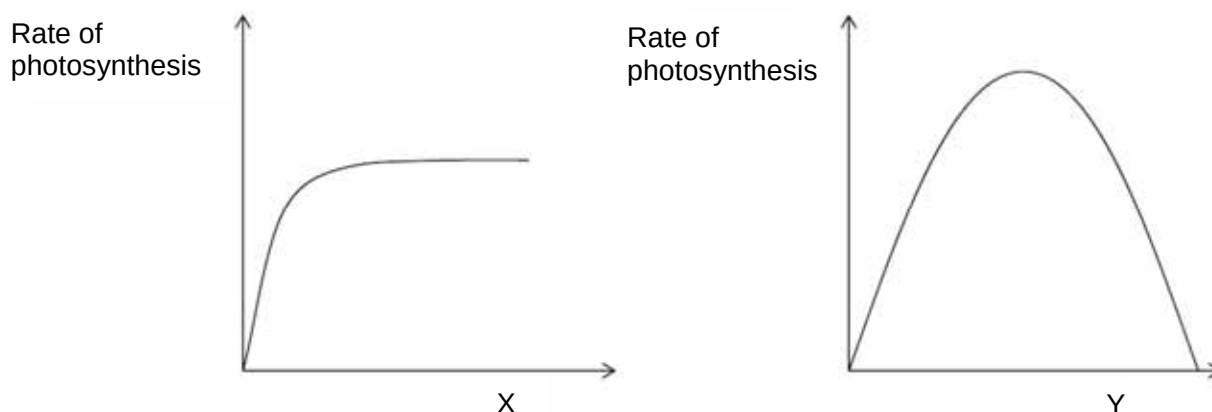
13. The diagram below shows the link reaction and stages of the Krebs cycle.

Which molecules are represented by the letters W, X, Y and Z?



	W	X	Y	Z
A	acetyl CoA	carbon dioxide	ADP	pyruvate
B	pyruvate	acetyl CoA	carbon dioxide	ADP
C	pyruvate	acetyl CoA	citrate	ATP synthase
D	acetyl CoA	pyruvate	carbon dioxide	ADP

14. Which limiting factors of photosynthesis are represented by X and Y?



	X	Y
A	carbon dioxide concentration	temperature
B	temperature	carbon dioxide concentration
C	light intensity	carbon dioxide concentration
D	temperature	light intensity

15. Purple buds of the morning glory flower, *Ipomoea*, open into blue flowers. As the flower opens, the pH of the vacuoles of the flower epidermal cells increases and this results in a change of colour from purple to blue.

A mutant purple-flowered morning glory plant carries recessive alleles of a gene **B/b**, coding for a membrane-bound ion pump, and is unable to increase the pH of the vacuole.

Both normal blue flowers and mutant purple flowers have the same anthocyanin pigment, coded by the dominant allele of the gene **A/a**. Plants with **aa** cannot produce anthocyanin and they have white flowers. The genes **A/a** and **B/b** are not linked.

A blue-flowered morning glory plant was crossed with a purple-flowered plant. Their offspring consisted of plants which are blue-flowered, purple-flowered as well as white-flowered.

What were the genotypes of the blue-flowered and purple-flowered plants?

	blue-flowered plant	purple-flowered plant
A	AABB	AaBb
B	AaBb	Aabb
C	aaBB	AaBB
D	aaBb	aaBb

16. A cross between a round-leaved, tall plant and round-leaved dwarf plant produced the following offspring:

121 round-leaved, tall plant
 124 round-leaved, dwarf plant
 42 oval-leaved, tall plant
 37 oval-leaved, dwarf plant

Key
R – round leaf
r – oval leaf
T – tall
t – dwarf

Which of the following statements are correct?

- I The parental genotypes are **RrTt** x **Rrtt**.
- II **R/r** and **T/t** loci are found on the same chromosome.
- III The progeny are results of a test cross.
- IV Crossing over occurred between the **R/r** and **T/t** loci during the formation of gametes in the parents.

- A I only
- B II and IV only
- C II and III only
- D II, III and IV only

17. When a polymerase chain reaction amplification was performed on a sample of human genomic DNA, multiple products of varying sizes were obtained, including one of the expected size.

Which of the following modifications to the protocol is the most likely to eliminate the extra PCR products?

- A raising the annealing temperature from 52°C to 58°C
- B raising the elongation temperature from 70°C to 76°C
- C decreasing the number of cycles from 30 to 20
- D decreasing the amount of template DNA from 1.0 µg to 0.5 µg

18. A restriction enzyme recognises specific DNA nucleotide sequences and cleaves the DNA at those positions. This can be used to detect DNA polymorphisms that are found within these sites.

Why is it not always possible to use restriction fragment length polymorphism (RFLP) to detect a mutation involving a disease-causing allele?

- A Not all disease-causing alleles have an associated RFLP.
 - B Point mutations can never be distinguished using RFLP.
 - C Use of RFLP is restricted to genetic diseases that are heritable.
 - D The temperature required in Southern hybridisation is unknown.
19. Which is a correct statement about obtaining human embryonic stem cells for research?
- A Removal of these cells is considered to be ethically acceptable as normal development of the embryo is not inhibited.
 - B The cells must be removed at an early stage of development from a region of the blastocyst known as the inner cell mass.
 - C The cells must be removed within a day following the successful fertilisation of the ovum by the sperm, and after checking for normal mitotic division.
 - D The region of the blastocyst from where the cells are removed is an area that develops at a later stage into the placenta.
20. Which of the following statements about the control of enzyme activity by phosphorylation is correct?
- A Phosphorylation of an enzyme is not a reversible process since it is a covalent modification.
 - B Phosphorylation of an enzyme occurs by protein phosphatases.
 - C Phosphorylation of an enzyme is an intracellular process and cannot occur in response to external signals.
 - D Phosphorylation of an enzyme results in a conformational change.
21. Which of the following statements about the integration of fat and carbohydrate metabolism control is correct?
- A Low insulin/glucagon ratio inhibits lipolysis in liver cell.
 - B High insulin/glucagon ratio increases the number of GLUT4 at the plasma membrane of pancreatic cell.
 - C Low insulin/glucagon ratio activates conversion of glycogen to glucose in liver cell.
 - D High insulin/glucagon ratio stimulates conversion amino acids to glucose in muscle cell.

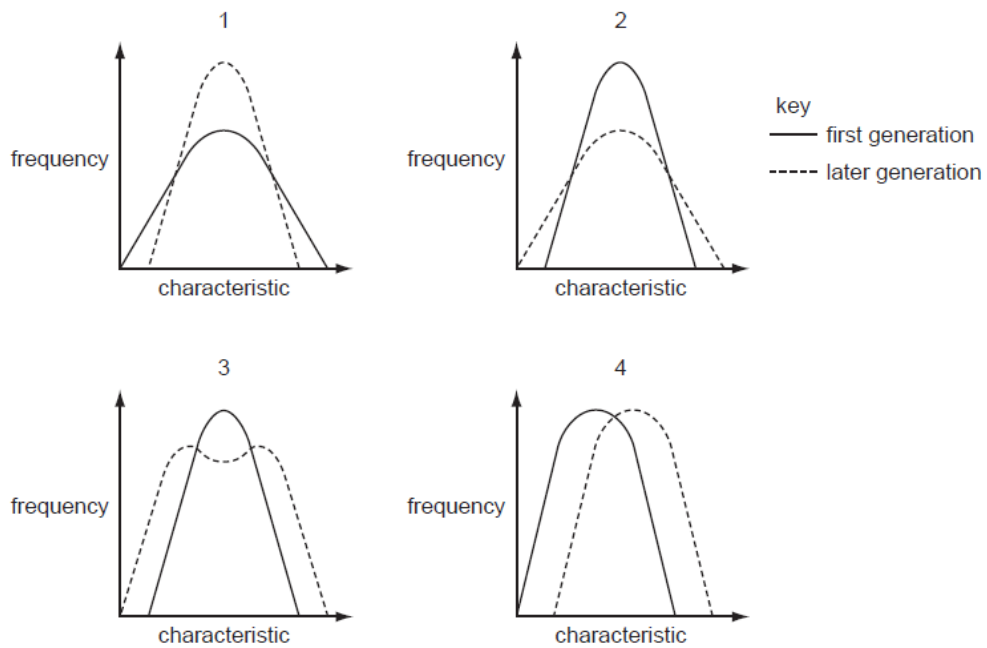
22. *Porphyrio hochstetteri*, is a flightless bird that is restricted to a small area of the South Island of New Zealand. It is one of only two remaining species of large, flightless, herbivorous birds from New Zealand. All the other species are extinct. *Porphyrio hochstetteri* was thought to be extinct, but a small population was discovered in 1948 in remote mountains of the South Island.

Which of the following statement(s) about the flightless bird population is/are incorrect?

	reasons for birds to evolve to become flightless in New Zealand	threats to remaining species of flightless birds in New Zealand
I	There were available niches.	A change in the environment.
II	There were no/few predators.	No change in the environment.
III	Suitable food on the ground was abundant.	The emergence of a disease.
IV	Flight requires considerable energy.	The introduction of predators by humans.

- A II only
 B I and III only
 C II and III only
 D All of the above

23. The graphs show frequency against a measured characteristic in the first and a later generation of a species.



Which graph represents each type of natural selection?

	directional	disruptive	stabilising
A	4	3	2
B	2	3	4
C	3	1	2
D	4	3	1

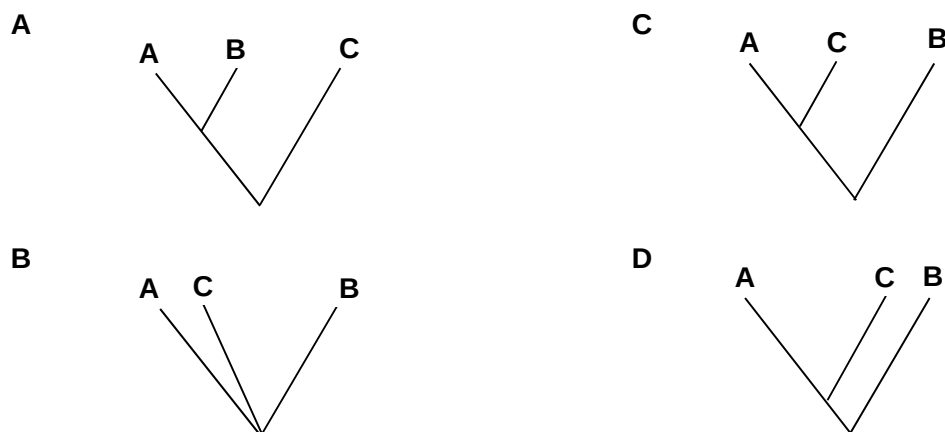
24. Phenylketonuria (PKU) is an autosomal recessive genetic disorder characterised by a deficiency in the enzyme phenylalanine hydroxylase. This enzyme is necessary to metabolise the amino acid phenylalanine to the amino acid tyrosine. When phenylalanine hydroxylase is deficient, phenylalanine accumulates and is converted into a toxic substance. Left untreated, this condition causes problems with brain development, leading to progressive mental retardation and seizures.

Which of the following maintains the presence of this harmful allele in a population's gene pool?

- A heterozygote advantage
 - B heterozygote protection
 - C frequency-dependent selection
 - D natural selection
25. Cytochrome c is a protein found in most organisms. The amino acid sequence of this protein varies between species. The number of differences in the amino acid sequences in cytochrome c between three species of chordates, A, B and C are shown in the table below.

	species B	species C
species A	11	3
species B		10

Based on this evidence, the phylogenetic tree that best represents the possible evolutionary relationships between the three species is:



26. What is/are the function(s) of macrophages during an immune response?

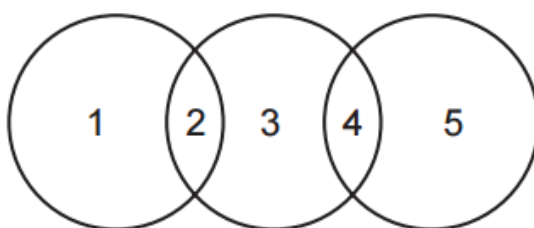
- 1 engulf pathogens and apoptotic cells
- 2 present antigens to naïve B cells
- 3 secrete signal molecules to recruit other immune cells

- A 1 only
- B 1 and 3 only
- C 2 and 3 only
- D All of the above

27. The heavy and light chains of an immunoglobulin are encoded by _____ that is/are rearranged during B cell development. This gives rise to a great diversity of immunoglobulins in an individual, where each immunoglobulin has a specific _____ complementary to an _____ on a pathogen.

- A a single continuous RNA sequence ; epitope ; antigen
- B a single continuous DNA sequence ; antigen-binding site ; antigenic determinant
- C sets of gene segments ; epitope ; antigen
- D sets of gene segments ; antigen binding site ; antigen

28. The diagram shows the relationship between influenza, tuberculosis and dengue.



Which of the following options is correct?

	1	2	3	4	5
A	dengue	bacteria	tuberculosis	airborne	influenza
B	influenza	waterborne	dengue	bacteria	tuberculosis
C	influenza	virus	tuberculosis	waterborne	dengue
D	tuberculosis	airborne	influenza	virus	dengue

29. Which effect of temperature increase on arctic ecosystems, will increase carbon dioxide in the atmosphere?
- A Greater production of plants due to warmer temperatures and changing vegetation.
 - B Greater decomposition of organic matter currently stored in permafrost.
 - C Less ice and snow will cause incoming radiation to be absorbed more readily.
 - D Melting ice from glaciers and icebergs will cause sea levels to rise.
30. Which statement is likely to be false?
- A Due to global warming, in the Northwest Atlantic, some of the commercial fish stocks will show significant range shifts due to warming sea temperatures.
 - B With increasing global temperatures, more radiation from sun will be reflected from the Earth's surface.
 - C Due to global warming, seedling survival of a few tree species will decrease due to lower rainfall in the Southern Appalachians mountains.
 - D Due to global warming, coastal mudflats and estuaries can disappear affecting the survival of migratory birds.

End of Paper

ANSWERS

1.	C	2.	D	3.	C	4.	D	5.	D
6.	B	7.	D	8.	C	9.	A	10.	A
11.	C	12.	C	13.	B	14.	A	15.	B
16.	A	17.	A	18.	A	19.	B	20.	D
21.	C	22.	A	23.	D	24.	B	25.	C
26.	B	27.	D	28.	D	29.	B	30.	B